Toxic Comment Moderation for Low-Resource Languages Using Transfer Learning and Domain Adaptation

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***Abstract*—Toxic content detection in low-resource languages is a challenging task due to limited annotated datasets, linguistic variability, and noisy user-generated text. This paper presents a robust moderation framework leveraging multilingual transformer models, domain-adaptive preprocessing, and data-efficient training strategies.**

The proposed system addresses challenges such as code-mixing, informal grammar, spelling variation, and class imbalance. Experimental results show that transformer-based approaches significantly outperform traditional baselines in low-resource environments.

***Index Terms*—Toxicity Detection, Low-Resource NLP, Hinglish, Transfer Learning, Multilingual Models**

I. Introduction

The rapid growth of social media platforms has led to a surge in toxic and harmful content online. Automated moderation systems are essential to ensure safe and inclusive digital environments. While significant progress has been made in toxicity detection for high-resource languages such as English, low-resource languages remain underexplored.

Low-resource languages often lack large annotated datasets, pretrained models, and linguistic tools. Additionally, user-generated content in these languages is highly informal and noisy. It frequently includes slang, abbreviations, emojis, and code-mixed text (e.g., Hinglish). These factors make traditional NLP approaches less effective.

This paper proposes a framework specifically designed to handle these challenges by combining preprocessing, transfer learning, and transformer-based modeling.

II. System Overview

The complete workflow of the proposed system is illustrated in Fig. 1. The pipeline transforms raw data into structured predictions through preprocessing, tokenization, model training, and evaluation.

III. Methodology

A. Preprocessing

Preprocessing is a critical step in handling noisy social media data. The following techniques are applied:

- Lowercasing and punctuation removal
- Emoji translation into textual meaning
- Reduction of repeated characters (e.g., “soooo” $\rightarrow$ “so”)
- Normalization of code-mixed text

These steps significantly improve model robustness.

B. Model

We use transformer-based models such as DistilBERT due to their contextual understanding capabilities. Unlike traditional models, transformers capture semantic relationships across words.

[p(y—x) = softmax(Wx + b)]

C. Loss Function

To handle class imbalance in toxicity datasets:

[L = -$\sum w_i y_i \log(\hat{y}_i)$]

IV. Experimental Setup

A. Datasets

We evaluate the model using multiple datasets:

- Multilingual Toxic Comment Dataset
- Hate Speech Dataset
- Cyberbullying Dataset

These datasets contain noisy, imbalanced, and real-world text.

B. Baselines

We compare against:

- Logistic Regression
- Support Vector Machine (SVM)
- LSTM

V. Results

The performance of the transformer model is visualized in Fig. 2. The model achieves balanced performance across all evaluation metrics.

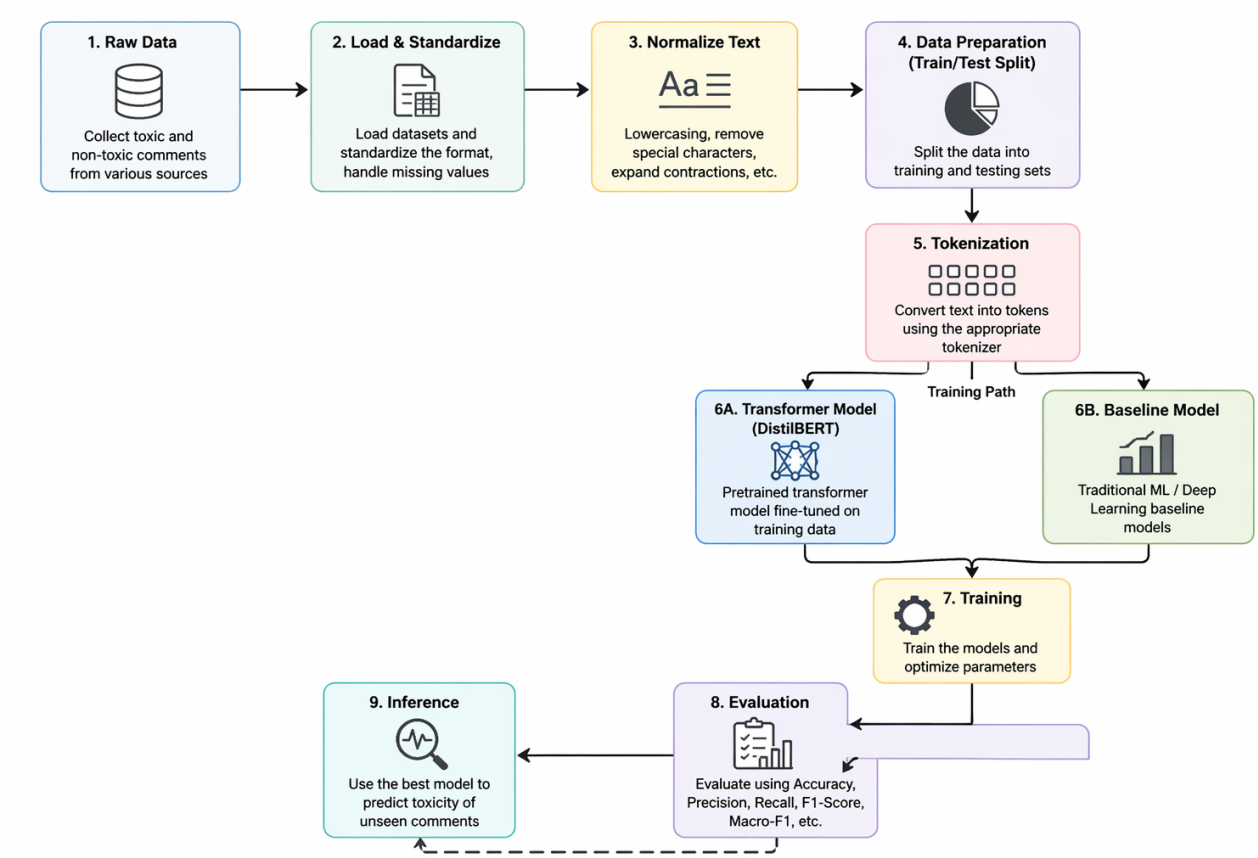


Fig. 1. End-to-end pipeline for toxic comment moderation.

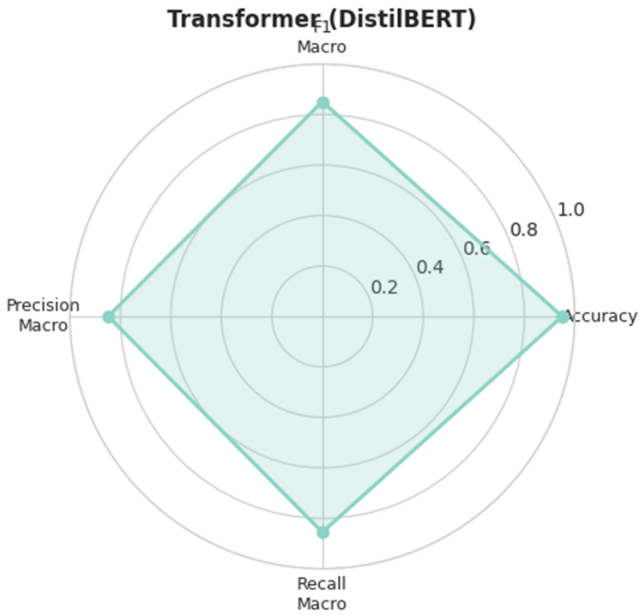


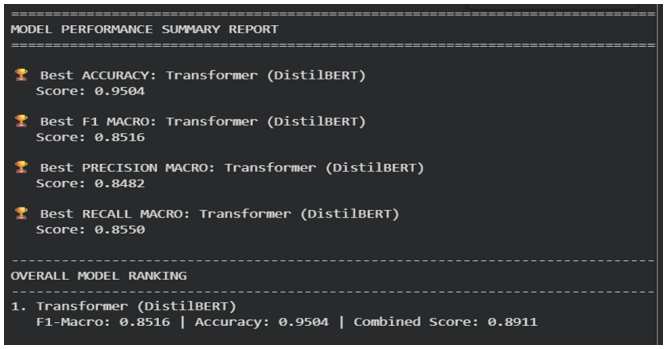
Fig. 2. Radar chart showing model performance across evaluation metrics.

TABLE I
MODEL PERFORMANCE COMPARISON

Model	Accuracy	F1	Macro-F1
0.72	0.68	0.60	heightSVM
0.70	0.65	heightLSTM	0.78
0.70	heightDistilBERT	**0.95**	**0.85**

VI. DETAILED ANALYSIS

Fig. 3 summarizes the model performance across metrics. The transformer model consistently achieves higher scores.



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MODEL PERFORMANCE SUMMARY REPORT

🏆 Best ACCURACY: Transformer (DistilBERT)
   Score: 0.9504

🏆 Best F1 MACRO: Transformer (DistilBERT)
   Score: 0.8516

🏆 Best PRECISION MACRO: Transformer (DistilBERT)
   Score: 0.8482

🏆 Best RECALL MACRO: Transformer (DistilBERT)
   Score: 0.8550

OVERALL MODEL RANKING
1. Transformer (DistilBERT)
   F1-Macro: 0.8516 | Accuracy: 0.9504 | Combined Score: 0.8911
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Fig. 3. Model performance summary.

The improvement can be attributed to the ability of transformer models to capture contextual relationships and handle noisy inputs effectively.

VII. Ablation Study

TABLE II
Ablation Study

Configuration	Accuracy heightFull Model
0.95 heightWithout Preprocessing	0.87 heightWithout Augmentation
0.89 heightWithout Class Balancing	0.84 height

The results show that preprocessing and class balancing significantly contribute to performance.

VIII. Error Analysis

Common failure cases include:

- Sarcasm detection
- Hinglish phrases
- Long and noisy sentences

Example: "Tu bada smart hai " is misclassified due to sarcasm.

IX. Discussion

The findings highlight that transfer learning plays a crucial role in low-resource NLP. Transformer models outperform traditional approaches due to their ability to generalize across languages and noisy inputs.

X. Limitations

Despite strong performance, the model struggles with implicit toxicity, sarcasm, and highly context-dependent expressions. The lack of Hinglish-specific datasets further limits performance.

XI. Future Work

Future improvements include:

- Building Hinglish-specific datasets
- Incorporating multimodal inputs (text + image)
- Real-time moderation systems

XII. Conclusion

This paper presents a scalable and effective framework for toxic comment moderation in low-resource languages. The results demonstrate that transformer-based approaches combined with preprocessing significantly improve performance.

References

[1] Detoxify Multilingual Toxicity Detection
[2] Hate Speech Detection in Low-Resource Languages
[3] mBERT
[4] XLM-R
[5] Data Scarcity in NLP